



Micro-Project Report

Institute Code: 0078

Academic Year: 2022-23

Program: Computer Technology

Course: Programming With Python(PWP)

Course Code:22616 **Scheme:** I

Semester: 6

Class: TYCM-I

Date of Report:10/04/2023

Title of Micro-Project: GUI based Grocery Billing System.

1.0 Problem Definition: GUI-based Grocery Billing System involves designing and implementing a user-friendly graphical user interface (GUI) that allows shopkeeper to make bill of sold item from a grocery store. The system includes features such as a list of available grocery products, an interface which asks user quantity of grocery product which they want, a calculator for computing the total cost of the purchased items, and a receipt generator. The system also includes features such as the ability to add new products to the list, remove products.

2.0 Rationale: A GUI-based Grocery Billing System can be useful to manage the grocery store the billing process of a grocery store. This system allows the shopkeeper to easily generate invoices of purchased grocery items. Here are some reasons why a GUI-based Grocery Billing System can be a good choice:

User-Friendly Interface: With a graphical user interface (GUI), the system can provide an intuitive and easy-to-use interface. This can help minimize training time and improve the overall user experience.

Improved Efficiency: A GUI-based system can automate many repetitive tasks, such as item scanning, calculation of the total cost, and printing receipts. This can help reduce errors and save time for the employees.

3.0 Aim /Benefits of Micro-Project:

- **Easy to use:** A GUI-based system makes it easy for users to interact with the software using visual elements.
- **Time-saving:** With a GUI-based system, users can quickly navigate through the various functions and features without having to remember complex commands or syntax. This can save a lot of time, especially for users.
- **Improved accuracy:** A GUI-based system can reduce the risk of errors as users can see what they are selecting or inputting on the screen. This can help to eliminate mistakes that we did physically calculating.

4.0 Course Outcomes Achieved (COs):

- a) Develop Python program to demonstrate use of Operators
- b) Perform operations on data structures in Python
- c) Develop functions for given problem.

5.0 Literature Review:

This project aims to learn the concept of GUI programming in python. A day's advancement in technology has become a big challenge in the field of marketing. Billing system was designed to provide high efficiency. The digital way to billing of sold product is much efficient than providing handwritten bill. This system will help to save the time and efforts of the shopkeeper. By providing user-friendly interface anyone can be handle this system easily. A GUI-based Grocery Billing System is an application that is used for billing customers in a grocery store or supermarket. It allows the user to enter the items purchased by the customer and calculates the total cost of the items. Python is a popular programming language used for developing such applications due to its simplicity and ease of use.

6.0 Actual Methodology followed:

- a) Finalizing the project topic i.e. GUI-based Grocery Billing System.
- b) Prepared proposal of project.
- c) Submit the proposal.
- d) Distributed the work among team members.
- e) Interpret and Filter the information collected.
- f) Prepared the Project report.
- g) Submitted the report to Subject Teacher.
- h) Submitted the whole project (hard, soft copy) along with the viva.

7.0 Actual Resources used:

Sr. No.	Name of Resource/material	Specifications	Qty	Remarks
1	Computer system	Hp intel i5 Processor, 8 GB Ram. 512 GB SSD	01	Available
2	Operating System	Windows 11	01	Available
3	Software	Chrome, Microsoft Word	01	Available
4	Websites	https://www.geeksforgeeks.org/python-simple-gui-calculator-using-tkinter/?ref=lbp	02	Available

8.0 Outputs of the Micro-Project:

Code:

```
from tkinter import *
import random
```

```
class Bill_App:
    def __init__(self, root):
        self.root = root
        self.root.geometry("1300x700+0+0")
        self.root.maxsize(width=1280, height=750)
        self.root.minsize(width=1280, height=750)
        self.root.title("Panchkrushna Kirana")
```

```

# *****Variables*****#
self.cus_name = StringVar()
self.c_phone = StringVar()
# For Generating Random Bill Numbers
x = random.randint(1, 9999)
self.c_bill_no = StringVar()
# Setting Value to variable
self.c_bill_no.set(str(x))

self.bath_soap = IntVar()
self.tea_powder = IntVar()
self.Coffee = IntVar()
self.dry_coconut = IntVar()
self.mango_pickle = IntVar()
self.rice = IntVar()
self.daal = IntVar()
self.food_oil = IntVar()
self.wheat = IntVar()
self.sugar = IntVar()
self.Milk = IntVar()
self.Salt = IntVar()
self.CoconutOil = IntVar()
self.groundnuts = IntVar()
self.biscuits = IntVar()
self.total_ = StringVar()
self.tax_cos = StringVar()

# *****Header of Application*****#
bg_color = "#FFF86D"
fg_color = "black"
lbl_color = 'black'
# Title of App
title = Label(self.root, text="Panchkrushna Kirana Store", bd=12, relief=GROOVE, fg=fg_color,
bg=bg_color,font=("times new roman", 30, "bold"), pady=3).pack(fill=X)

# *****Customers Frame*****#
F1 = LabelFrame(text="Customer Details", font=("times new roman", 12, "bold"), fg="red",
bg=bg_color,relief=GROOVE, bd=10)
F1.place(x=0, y=80, relwidth=1)

# *****Customer Name*****#
cname_lbl = Label(F1, text="Customer Name", bg=bg_color, fg=fg_color,font=("times new roman",
15, "bold")).grid(row=0, column=0, padx=10, pady=5)
cname_en = Entry(F1, bd=8, relief=GROOVE, textvariable=self.cus_name)
cname_en.grid(row=0, column=1, ipady=4, ipadx=30, pady=5)

```

```

#####Customer Phone#####
cphon_lbl = Label(F1, text="Phone No", bg=bg_color, fg=fg_color, font=("times new roman", 15,
"bold")).grid(row=0, column=2, padx=20)
cphon_en = Entry(F1, bd=8, relief=GROOVE, textvariable=self.c_phone)
cphon_en.grid(row=0, column=3, ipady=4, ipadx=30, pady=5)

# #####Customer Bill No#####
cbill_lbl = Label(F1, text="Bill No.", bg=bg_color, fg=fg_color, font=("times new roman", 15,
"bold"))
cbill_lbl.grid(row=0, column=4, padx=20)
cbill_en = Entry(F1, bd=8, relief=GROOVE, textvariable=self.c_bill_no)
cbill_en.grid(row=0, column=5, ipadx=30, ipady=4, pady=5)

# ##### Frame#####
F2 = LabelFrame(self.root, bd=10, relief=GROOVE, bg=bg_color, fg="gold", font=("times new
roman", 13, "bold"))
F2.place(x=5, y=180, width=325, height=380)

# #####Frame Content#####
bath_lbl = Label(F2, font=("times new roman", 15, "bold"), fg=lbl_color, bg=bg_color, text="Bath
Soap")
bath_lbl.grid(row=0, column=0, padx=10, pady=20)
bath_en = Entry(F2, bd=8, relief=GROOVE, textvariable=self.bath_soap)
bath_en.grid(row=0, column=1, ipady=5, ipadx=5)

#####Tea Powder#####
face_lbl = Label(F2, font=("times new roman", 15, "bold"), fg=lbl_color, bg=bg_color, text="Tea
Powder")
face_lbl.grid(row=1, column=0, padx=10, pady=20)
face_en = Entry(F2, bd=8, relief=GROOVE, textvariable=self.tea_powder)
face_en.grid(row=1, column=1, ipady=5, ipadx=5)

# #####Coffee#####
wash_lbl = Label(F2, font=("times new roman", 15, "bold"), fg=lbl_color, bg=bg_color,
text="Coffee")
wash_lbl.grid(row=2, column=0, padx=10, pady=20)
wash_en = Entry(F2, bd=8, relief=GROOVE, textvariable=self.Coffee)
wash_en.grid(row=2, column=1, ipady=5, ipadx=5)

# #####Dry Coconut#####
hair_lbl = Label(F2, font=("times new roman", 15, "bold"), fg=lbl_color, bg=bg_color, text="Dry
Coconut")
hair_lbl.grid(row=3, column=0, padx=10, pady=20)
hair_en = Entry(F2, bd=8, relief=GROOVE, textvariable=self.dry_coconut)
hair_en.grid(row=3, column=1, ipady=5, ipadx=5)

#####Mango Pickle#####

```

```

lot_lbl = Label(F2, font=("times new roman", 15, "bold"), fg=lbl_color, bg=bg_color, text="Mango Pickle")
lot_lbl.grid(row=4, column=0, padx=10, pady=20)
lot_en = Entry(F2, bd=8, relief=GROOVE, textvariable=self.mango_pickle)
lot_en.grid(row=4, column=1, ipady=5, ipadx=5)

# *****Grocery Frame*****#
F2 = LabelFrame(self.root, bd=10, relief=GROOVE, bg=bg_color, fg="gold", font=("times new roman", 13, "bold"))
F2.place(x=330, y=180, width=325, height=380)

# *****Frame Content
rice_lbl = Label(F2, font=("times new roman", 15, "bold"), fg=lbl_color, bg=bg_color, text="Rice")
rice_lbl.grid(row=0, column=0, padx=10, pady=20)
rice_en = Entry(F2, bd=8, relief=GROOVE, textvariable=self.rice)
rice_en.grid(row=0, column=1, ipady=5, ipadx=5)

# *****Food Oil
oil_lbl = Label(F2, font=("times new roman", 15, "bold"), fg=lbl_color, bg=bg_color, text="Food Oil")
oil_lbl.grid(row=1, column=0, padx=10, pady=20)
oil_en = Entry(F2, bd=8, relief=GROOVE, textvariable=self.food_oil)
oil_en.grid(row=1, column=1, ipady=5, ipadx=5)

# *****Daal
daal_lbl = Label(F2, font=("times new roman", 15, "bold"), fg=lbl_color, bg=bg_color, text="Daal")
daal_lbl.grid(row=2, column=0, padx=10, pady=20)
daal_en = Entry(F2, bd=8, relief=GROOVE, textvariable=self.daal)
daal_en.grid(row=2, column=1, ipady=5, ipadx=5)

#*****Wheat
wheat_lbl = Label(F2, font=("times new roman", 15, "bold"), fg=lbl_color, bg=bg_color, text="Wheat")
wheat_lbl.grid(row=3, column=0, padx=10, pady=20)
wheat_en = Entry(F2, bd=8, relief=GROOVE, textvariable=self.wheat)
wheat_en.grid(row=3, column=1, ipady=5, ipadx=5)

# =====Sugar
sugar_lbl = Label(F2, font=("times new roman", 15, "bold"), fg=lbl_color, bg=bg_color, text="Sugar")
sugar_lbl.grid(row=4, column=0, padx=10, pady=20)
sugar_en = Entry(F2, bd=8, relief=GROOVE, textvariable=self.sugar)
sugar_en.grid(row=4, column=1, ipady=5, ipadx=5)

F2 = LabelFrame(self.root, bd=10, relief=GROOVE, bg=bg_color, fg="gold", font=("times new roman", 13, "bold"))
F2.place(x=655, y=180, width=325, height=380)

```

```

# =====Frame Content
Milk_lbl = Label(F2, font=("times new roman", 15, "bold"), fg=lbl_color, bg=bg_color,
text="Milk")
Milk_lbl.grid(row=0, column=0, padx=10, pady=20)
Milk_en = Entry(F2, bd=8, relief=GROOVE, textvariable=self.Milk)
Milk_en.grid(row=0, column=1, ipady=5, ipadx=5)

# =====Salt
cock_lbl = Label(F2, font=("times new roman", 15, "bold"), fg=lbl_color, bg=bg_color, text="Salt")
cock_lbl.grid(row=1, column=0, padx=10, pady=20)
cock_en = Entry(F2, bd=8, relief=GROOVE, textvariable=self.Salt)
cock_en.grid(row=1, column=1, ipady=5, ipadx=5)

# =====Coconut Oil
CoconutOil_lbl = Label(F2, font=("times new roman", 15, "bold"), fg=lbl_color, bg=bg_color,
text="Coconut Oil")
CoconutOil_lbl.grid(row=2, column=0, padx=10, pady=20)
CoconutOil_en = Entry(F2, bd=8, relief=GROOVE, textvariable=self.CoconutOil)
CoconutOil_en.grid(row=2, column=1, ipady=5, ipadx=5)

# =====Groundnuts
cold_lbl = Label(F2, font=("times new roman", 15, "bold"), fg=lbl_color, bg=bg_color,
text="Groundnuts ")
cold_lbl.grid(row=3, column=0, padx=10, pady=20)
cold_en = Entry(F2, bd=8, relief=GROOVE, textvariable=self.groundnuts)
cold_en.grid(row=3, column=1, ipady=5, ipadx=5)

# =====Biscuits
bis_lbl = Label(F2, font=("times new roman", 15, "bold"), fg=lbl_color, bg=bg_color,
text="Biscuits")
bis_lbl.grid(row=4, column=0, padx=10, pady=20)
bis_en = Entry(F2, bd=8, relief=GROOVE, textvariable=self.biscuits)
bis_en.grid(row=4, column=1, ipady=5, ipadx=5)

# =====Bill Area=====#
F3 = Label(self.root, bd=10, relief=GROOVE)
F3.place(x=960, y=180, width=325, height=380)
# =====
bill_title = Label(F3, text="Bill Area", font=("Lucida", 13, "bold"), bd=7, relief=GROOVE)
bill_title.pack(fill=X)

# =====
scroll_y = Scrollbar(F3, orient=VERTICAL)
self.txt = Text(F3, yscrollcommand=scroll_y.set)
scroll_y.pack(side=RIGHT, fill=Y)
scroll_y.config(command=self.txt.yview)

```

```

self.txt.pack(fill=BOTH, expand=1)

# =====Buttons Frame=====
F4 = LabelFrame(self.root, text='Bill Menu', bd=10, relief=GROOVE, bg=bg_color, fg="red",
                font=("times new roman", 13, "bold"))
F4.place(x=0, y=560, relwidth=1, height=100)

# =====
lbl = Label(F4, font=("times new roman", 15, "bold"), fg=lbl_color, bg=bg_color, text="Total ")
lbl.grid(row=0, column=0, padx=10, pady=0)
cosm_en = Entry(F4, bd=8, relief=GROOVE, textvariable=self.total_)
cosm_en.grid(row=0, column=1, ipady=2, ipadx=5)

# =====Tax
cosmt_lbl = Label(F4, font=("times new roman", 15, "bold"), fg=lbl_color, bg=bg_color, text="
GST")
cosmt_lbl.grid(row=0, column=2, padx=30, pady=0)
cosmt_en = Entry(F4, bd=8, relief=GROOVE, textvariable=self.tax_cos)
cosmt_en.grid(row=0, column=3, ipady=2, ipadx=5)
# =====
total_btn = Button(F4, text="Total", bg=bg_color, fg=fg_color, font=("lucida", 12, "bold"), bd=7,
relief=GROOVE,command=self.total)
total_btn.grid(row=0, column=4, ipadx=20, padx=30)

# =====
genbill_btn = Button(F4, text="Generate Bill", bg=bg_color, fg=fg_color, font=("lucida", 12,
"bold"), bd=7,
                    relief=GROOVE, command=self.bill_area)
genbill_btn.grid(row=0, column=5, ipadx=20)

# =====
clear_btn = Button(F4, text="Clear", bg=bg_color, fg=fg_color, font=("lucida", 12, "bold"), bd=7,
relief=GROOVE,
                    command=self.clear)
clear_btn.grid(row=0, column=6, ipadx=20, padx=30)

# =====
exit_btn = Button(F4, text="Exit", bg=bg_color, fg=fg_color, font=("lucida", 12, "bold"), bd=7,
relief=GROOVE,
                    command=self.exit)
exit_btn.grid(row=0, column=7, ipadx=20)

# Function to get total prices
def total(self):
    # =====Total Prices
    self.total__prices = (
        (self.bath_soap.get() * 20) +

```

```

        (self.tea_powder.get() * 100) +
        (self.Coffee.get() * 140) +
        (self.dry_coconut.get() * 240) +
        (self.mango_pickle.get() * 260)+
        (self.wheat.get() * 100)+
        (self.food_oil.get() * 180) +
        (self.daal.get() * 80) +
        (self.rice.get() * 35) +
        (self.sugar.get() * 45) +
        (self.Milk.get() * 70) +
        (self.CoconutOil.get() * 50) +
        (self.Salt.get() * 10) +
        (self.groundnuts.get() * 50) +
        (self.biscuits.get() * 20)
    )
    self.total_.set("Rs. " + str(self.total__prices))
    self.tax_cos.set("Rs. " + str(float(self.total__prices * 0.05)))

```

Function For Text Area

```

def welcome_soft(self):
    self.txt.delete('1.0', END)
    self.txt.insert(END, "    Panchkrushna Kirana Store\n")
    self.txt.insert(END, f"\nBill No. : {str(self.c_bill_no.get())}")
    self.txt.insert(END, f"\nCustomer Name : {str(self.cus_name.get())}")
    self.txt.insert(END, f"\nPhone No. : {str(self.c_phone.get())}")
    self.txt.insert(END, "\n=====")
    self.txt.insert(END, "\nProduct      Qty      Price")
    self.txt.insert(END, "\n=====")

```

Function to clear the bill area

```

def clear(self):
    self.txt.delete('1.0', END)

```

Add Product name , qty and price to bill area

```

def bill_area(self):
    self.welcome_soft()
    if self.bath_soap.get() != 0:
        self.txt.insert(END, f"\nBath Soap      {self.bath_soap.get()}      {self.bath_soap.get() * 20}")
    if self.tea_powder.get() != 0:
        self.txt.insert(END, f"\nTea Powder      {self.tea_powder.get()}      {self.tea_powder.get() *
100}")
    if self.Coffee.get() != 0:
        self.txt.insert(END, f"\nCoffee      {self.Coffee.get()}      {self.Coffee.get() * 140}")
    if self.dry_coconut.get() != 0:
        self.txt.insert(END, f"\nDry Coconut      {self.dry_coconut.get()}      {self.dry_coconut.get() *
240}")

```



```

        if self.mango_pickle.get() != 0:
            self.txt.insert(END, f"\nMango Pickle      {self.mango_pickle.get()}
{self.mango_pickle.get() * 260}")
        if self.wheat.get() != 0:
            self.txt.insert(END, f"\nWheat          {self.wheat.get()}          {self.wheat.get() * 100}")
        if self.food_oil.get() != 0:
            self.txt.insert(END, f"\nFood Oil      {self.food_oil.get()}          {self.food_oil.get() * 180}")
        if self.daal.get() != 0:
            self.txt.insert(END, f"\nDaal          {self.daal.get()}          {self.daal.get() * 80}")
        if self.rice.get() != 0:
            self.txt.insert(END, f"\nRice          {self.rice.get()}          {self.rice.get() * 35}")
        if self.sugar.get() != 0:
            self.txt.insert(END, f"\nSugar          {self.sugar.get()}          {self.sugar.get() * 45}")
        if self.Milk.get() != 0:
            self.txt.insert(END, f"\nMilk          {self.Milk.get()}          {self.Milk.get() * 70}")
        if self.CoconutOil.get() != 0:
            self.txt.insert(END, f"\nCoconut Oil    {self.CoconutOil.get()}          {self.CoconutOil.get() *
50}")
        if self.Salt.get() != 0:
            self.txt.insert(END, f"\nSalt          {self.Salt.get()}          {self.Salt.get() * 10}")
        if self.groundnuts.get() != 0:
            self.txt.insert(END, f"\nGroundnuts     {self.groundnuts.get()}          {self.groundnuts.get() *
50}")
        if self.biscuits.get() != 0:
            self.txt.insert(END, f"\nBiscuits        {self.biscuits.get()}          {self.biscuits.get() * 20}")
        self.txt.insert(END, "\n=====")
        self.txt.insert(END, f"\nTotal : {self.total__prices + self.total__prices * 0.05}")
        self.txt.insert(END, "\nThank You...! Visit Again")
        self.config(state=DISABLED)
        self.print()

```

Function to exit

```

def exit(self):
    self.root.destroy()

```

Function To Clear All Fields

```

root = Tk()
object = Bill_App(root)
root.mainloop()

```

Result:

Panchkrushna Kirana

Panchkrushna Kirana Store

Customer Details

Customer Name

Mr.Monu Ugale

Phone No

7620436897

Bill No.

1609

Bath Soap

01

Tea Powder

02

Coffee

03

Dry Coconut

01

Mango Pickle

01

Rice

02

Food Oil

03

Daal

04

Wheat

05

Sugar

01

Milk

01

Salt

02

Coconut Oil

05

Groundnuts

04

Biscuits

06

Bill Area

Panchkrushna Kirana Store

Bill No. : 1609

Customer Name : Mr.Monu Ugale

Phone No. : 7620436897

Product	Qty	Price
Bath Soap	1	20
Tea Powder	2	200
Coffee	3	420
Dry Coconut	1	240
Mango Pickle	1	260
Wheat	5	500
Food Oil	3	540
Daal	4	320
Rice	2	70
Sugar	1	45
Milk	1	70
Coconut Oil	5	250

Bill Menu

Total

Rs. 3275

GST

Rs. 163.75

Total

Generate Bill

Clear

Exit

Panchkrushna Kirana

Panchkrushna Kirana Store

Customer Details

Customer Name

Mr.Monu Ugale

Phone No

7620436897

Bill No.

1609

Bath Soap

01

Tea Powder

02

Coffee

03

Dry Coconut

01

Mango Pickle

01

Rice

02

Food Oil

03

Daal

04

Wheat

05

Sugar

01

Milk

01

Salt

02

Coconut Oil

05

Groundnuts

04

Biscuits

06

Bill Area

Product	Qty	Price
Bath Soap	1	20
Tea Powder	2	200
Coffee	3	420
Dry Coconut	1	240
Mango Pickle	1	260
Wheat	5	500
Food Oil	3	540
Daal	4	320
Rice	2	70
Sugar	1	45
Milk	1	70
Coconut Oil	5	250
Salt	2	20
Groundnuts	4	200
Biscuits	6	120

Total : 3438.75

Thank You...! Visit Again

Bill Menu

Total

Rs. 3275

GST

Rs. 163.75

Total

Generate Bill

Clear

Exit

9.0 Skill Developed / Learning outcome of this Micro-Project:

9.1 Soft Skills:

- Problem Solving
- Leadership
- Teamwork
- Time Management
- Decision Making

9.2 Technical Skills:

- Software Efficiency
- Word Processing
- Knowledge of performing Technical Tasks

10.0 Applications of Micro Project:

- 1) To study of Billing System.
- 2) To gain basic information about prices of Grocery System.
- 3) It is used to manage Grocery Store.

11.0 Name of Group Members:

Enrolment	Roll No.	Seat No.	Name of Students	Student Signature
2100780748	69	386408	Mohan Suresh Ugale	
2100780745	67	386406	Amit Keshav More	
2000780249	52	386271	Shubham Harendra Ram	
2000780240	44	386263	Anup Jitendra Pardeshi	

Date: 10 / 04/2023

Evaluated by: Dated Signature of Guide: _____

Name of Guide: Mrs. M. P. Bhosale